Mathematical expectation, variance for two dimensional random variables

Let X and Y be random variables with joint probability distribution. The mean (or) expected value of the random variable $Z = g(X, Y)$ is

$$E(Z) = \sum_{X=x} \sum_{Y=y} g(X,Y)\, p(x,y) \quad \text{if } X, Y \text{ are discrete}$$

$$E(Z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(X,Y)\, f(x,y)\, dx\, dy \quad \text{if } X, Y \text{ are continuous}$$

Note:-

If $g(X,Y) = X$ then

$$E(X) = \sum_{x} \sum_{y} x\, p(x,y) = \sum_{x} x\, P_X(x) \quad \text{for } X \text{ discrete}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x\, f(x,y)\, dx\, dy = \int_{-\infty}^{\infty} x\, f_X(x)\, dx \quad \text{for } X \text{ continuous}$$

2) If $g(X,Y)=Y$ then

$$E(Y)=\sum_{x}\sum_{y} y\,p(x,y)=\sum_{y} y\,P_Y(y) \quad ; Y \text{ is discrete}$$

$$=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} y\,f(x,y)\,dx\,dy=\int_{-\infty}^{\infty} y\,f_Y(y)\,dy$$

Y is continuous.

3) If $g(X,Y)=XY$ then

$$E(XY)=\sum_{x}\sum_{y} xy\,p(x,y) \quad ; X, Y \text{ are discrete}$$

$$=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\,f(x,y)\,dx\,dy \quad X, Y \text{ are continuous.}$$

4) If X, Y are statistically independent then

$$E(XY)=E(X)\cdot E(Y).$$

Covariance and correlation co-efficient: Let X and Y be two random variables with joint probability distribution then

$$\operatorname{cov}(X,Y) = E(XY) - E(X)\cdot E(Y)$$

Correlation between X and Y is $\rho(X,Y) = \dfrac{\operatorname{cov}(X,Y)}{\sigma_X \sigma_Y}$

where σ_X - is the standard deviation of X.
σ_Y - is the standard deviation of Y.

[Variance $(X) = E(X^2) - [E(X)]^2$; Standard deviation $= \sqrt{\text{Variance}}$]

Pb Let X and Y be the random variable with joint probability mass function:

$p(x,y)$		$x=0$	$x=1$	$x=2$
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0
	2	$\frac{1}{28}$	0	0

Find $E(X)$, $E(Y)$, $E(XY)$, $\mathrm{cov}(X,Y)$

Soln:-

$p(x,y)$		$x=0$	$x=1$	$x=2$	$P_Y(y)$
y	0	$3/28$	$9/28$	$3/28$	$15/28$
	1	$3/14$	$3/14$	0	$12/28$
	2	$1/28$	0	0	$1/28$
$P_X(x)$		$10/28$	$15/28$	$3/28$	1

$$\therefore E(X) = \sum_x x\, P_X(x) = 0 \cdot P_X(0) + 1 \cdot P_X(1) + 2 \cdot P_X(2)$$
$$= 0 \cdot \frac{10}{28} + 1 \cdot 15/28 + 2 \cdot 3/28$$
$$= 21/28.$$

$$E(Y) = \sum_y y\, P_Y(y) = 0 \cdot P_Y(0) + 1 \cdot P_Y(1) + 2 \cdot P_Y(2)$$
$$= 0 \cdot 15/28 + 1 \cdot \frac{12}{28} + 2 \cdot \frac{1}{28}$$
$$= 14/28.$$

$$E(XY) = \sum_x \sum_y xy\, p(x,y) = \sum_{x=0}^{2} \sum_{y=0}^{2} xy\, p(x,y)$$

$$= 0\cdot 0\, p(0,0) + 0\cdot 1\, p(0,1) + 0\cdot 2\, p(0,2)$$

$$1\cdot 0\, p(1,0) + 1\cdot 1\, p(1,1) + 1\cdot 2\, p(1,2)$$

$$2\cdot 0\, p(2,0) + 2\cdot 1\, p(2,1) + 2\cdot 2\, p(2,2)$$

$$= 0 + 0 + 0 + 0 + p(1,1) + 2p(1,2) + 0 + 2p(2,1) + 4p(2,2)$$

$$= \frac{3}{14} + 2\cdot 0 + 2\cdot 0 + 4\cdot 0$$

$$= 3/14$$

$$\mathrm{Cov}(X,Y) = E(XY) - E(X) - E(Y) = \frac{3}{14} - \frac{21}{28}\cdot\frac{14}{28}$$

$$= -\frac{1}{6}.$$

Pb

Find $E\left(\frac{Y}{X}\right)$ for the density function

$$f(x,y) = \begin{cases} \dfrac{x(1+3y^2)}{4}, & 0<x<2,\ 0<y<1 \\ 0 & \text{elsewhere} \end{cases}$$

soln

$$E\left(\frac{Y}{X}\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \frac{y}{x} f(x,y)\,dx\,dy = \int_0^1\int_0^2 \frac{y}{x}\,\frac{x(1+3y^2)}{4}\,dx\,dy$$

$$= \int_0^1\int_0^2 \frac{y(1+3y^2)}{4}\,dx\,dy$$

$$= \int_0^1 \frac{y(1+3y^2)}{2}\,dy = 5/8$$

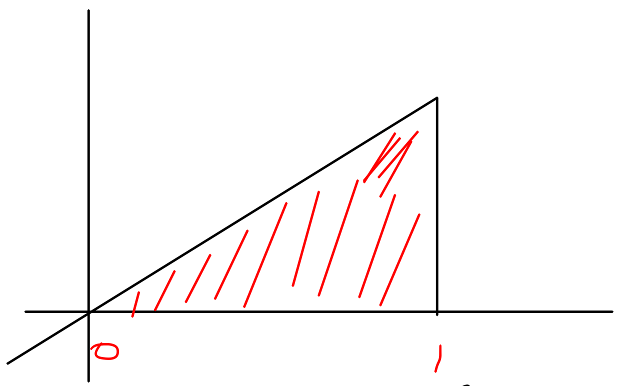
Pb The Joint density function of the continuous random variables X and Y is

$$f(x,y) = \begin{cases} 8xy & 0 \le y \le x \le 1 \\ 0 & \text{elsewhere} \end{cases}$$

1) Find the Covariance of X and Y.

2) Find the correlation between X and Y.

Soln:-



$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_0^x 8xy\,dy = 8x\left[\frac{y^2}{2}\right]_0^x$$

$$= 4x^3 \quad 0 \le x \le 1.$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_y^1 8xy\,dx = 8y\left[\frac{x^2}{2}\right]_y^1$$

$$= 4y(1-y^2) \quad 0 \le y \le 1$$

$$E(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_0^1 x\,4x^3\,dx = 4/5$$

$$E(Y) = \int_{-\infty}^{\infty} y f_Y(y)\,dy = \int_0^1 y\,4y(1-y^2)\,dy = \frac{8}{15}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx = \int_0^1 x^2\,4x^3\,dx = 2/3$$

$$E(Y^2) = \int_{-\infty}^{\infty} y^2 f_Y(y)\,dy = \int_0^1 y^2\,4y(1-y^2)\,dy = 1/3$$

$$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f(x,y)\,dx\,dy = \int_0^1\int_y^1 xy\,8xy\,dx\,dy$$

$$= \int_0^1 8y^2\left[\frac{x^3}{3}\right]_y^1 dy = \frac{8}{3}\int_0^1 y^2(1-y^3)\,dy$$

$$= 4/9$$

$$\mathrm{cov}(X,Y) = E(XY) - E(X)E(Y)$$

$$= 4/9 - \left(\frac{4}{5}\right)\left(\frac{8}{15}\right) = \frac{4}{225}$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{2}{3} - \left[\frac{4}{5}\right]^2 = 2/75$$

$$\mathrm{Var}(Y) = E(Y^2) - [E(Y)]^2 = \frac{1}{3} - \left[\frac{8}{15}\right]^2 = 11/225$$

$$\therefore \sigma_X = \sqrt{2/75} \; ; \; \sigma_Y = \sqrt{\frac{11}{225}}$$

$$\rho(X,Y) = \frac{\mathrm{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\frac{4}{225}}{\sqrt{2/75}\sqrt{11/225}}$$

$$= 4/\sqrt{66}$$